PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

umhackathon@um.edu.my

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Email: umhackathon@um.edu.my

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Email: umhackathon@um.edu.my

Project Name: CloThinks

Version: 2.0

1. Project Overview

- **Problem Statement:** Many SMEs' clothing businesses struggle with inefficient inventory management, leading to overstocking, waste, and missed sales opportunities. These challenges arise from fragmented and unstructured data, limited access to analytical tools, and heavy reliance on intuition or past experience to make decisions effectively and timely.

  In addition, unsold clothing items are difficult to clear due to limited styling guidance and a lack of personalised recommendations for customers.

- **Target Domain:** AI-driven Inventory Intelligence for SME Fashion Retailers, focused on trend-aware inventory optimization, minimizing deadstock, and data-backed pricing strategies for the fashion and apparel sector.

- **Proposed Solution Summary:** CloThinks prompts an interface in simple English language to help SME Fashion Retailers improve inventory management and reduce waste by tracking stock in real time, identifying slow-moving items, and predicting demand. It also functions as an AI fashion stylist, generating outfit combinations by pairing slow moving items with items customers are likely to buy to help store owners turn unsold items into sellable looks. Hence, shopping experiences are increased by assisting customers with personalized style recommendations and accurate size matching.

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2. Background & Business Objective

- **Background of the Problem:** Many clothing store owners in Malaysia and throughout the world struggle to expand their businesses because they make important decisions based more on "gut feelings" or simple habits than on actual data. This is because their company information is frequently spread out among many notebooks or simple spreadsheets, making it difficult for them to see the big picture or make quick and strategic decisions. This reliance on manual work and intuition often leads to costly mistakes, such as ordering too much of low-selling clothes or missing out on a new trend because they didn't have the data to see it coming. Without a system that can really analyze and interpret complicated information, these businesses face difficulties when attempting to cut expenses or improve everyday sales. The fashion market changes fast, and staying ahead requires more than just simple automation; it requires "decision intelligence" that can think through different options and recommend the best path forward.

  Currently, many small retailers lack the tools to analyze both their sales numbers and messy information like customer reviews or fashion trends at the same time. This creates a gap where store owners cannot effectively plan their inventory to make the most profit. By using CloThink, these business owners can turn their fragmented data into clear, easy-to-understand insights that help them save time and make more money.

- **Importance of Solving This Issue:** By replacing slow, intuition-based processes with Z.AI-powered real-time intelligence, SME Fashion Retailers can drastically reduce capital trapped in deadstock and minimize textile waste. Ultimately, this empowers small business owners to compete with large corporations by enabling them to respond to market trends in seconds rather than days, transforming operational efficiency into measurable economic growth. Solving this issue is not just a business necessity but an environmental imperative, reducing the amount of unsold clothing that ends up in landfills.

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- **Strategic Fit / Impact:** CloThinks enables SME digital transformation by using AI for real-time inventory insights and faster decision-making. It increases inventory turnover, reduces deadstock, improves profit margins, and supports a sustainable fashion ecosystem by minimizing textile waste to enhance competitiveness at the same time.

3. Product Purpose

3.1. Main Goal of the System: To act as an AI-powered inventory & styling partner that helps clothing SMEs eliminate financial waste from unsold stock and unused fabric, while simultaneously increasing sales through intelligent size and style-based customer matching from existing inventory.

3.2. Intended Users (Target Audience):

- SME fashion retailers
- Store/ Inventory managers
- Production staff
- Online customers
- Small online fashion sellers

4. System Functionalities

4.1. Description: CloThinks is an AI-powered decision intelligence platform that transforms fragmented retail data into actionable insights for clothing SMEs. Through a simple natural language interface, the system enables real-time inventory tracking, demand prediction, and AI-driven outfit generation to optimize sales.

CloThinks is a system that functions as a comprehensive decision intelligence tool designed to empower clothing retailers by transforming fragmented data into strategic growth. Fundamentally, by watching stock in real time, identifying slow-moving items, and accurately predicting future demand, the platform enhances inventory management and minimizes waste. In addition to tracking, the system functions as an AI fashion stylist, it analyze unorganized data and generating outfit combinations from stock. By combining remaining, unsold things with bestselling pieces that shoppers are already likely to purchase, this creative strategy assists store

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managers in transforming them into new, profitable designs. By providing clear explanations for these recommendations, the system ensures that every decision is data-driven and aimed at optimizing outcomes such as cost savings and growth in revenue.

4.2. Key Functionalities:

- **Smart Inventory Intelligence**: Tracks stock in real time, analyzes sales pattern and identifies overstocked items.
- **Conversational Interface:** Allows users to retrieve suggestions and interact with the system using simple English prompts for easier understanding.
- **AI Outfit Matching Recommendation**: Generate outfits combinations from existing inventory to help convert unsold items into attractive and sellable bundles.
- **Demand Prediction:** Able to predict future demand trends and suggest solutions such as restocking and bundling.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: We chose Z.AI GLM because it transforms the system from a passive tracker into a decision-making partner. While traditional tools can record figures, Z.AI's GLM is unique in that it can evaluate both organized sales data and unstructured fashion trends to provide context-aware reasoning. This capability is essential for the system's "AI Fashion Stylist" function, as it allows the model to understand the aesthetic relationships between garments—pairing slow-moving items with trending items in a way that feels natural and appealing to customers. Furthermore, the GLM provides a level of decision intelligence that goes beyond simple automation by offering clear, human-understandable explanations for its strategies. Instead of just flagging an item as "slow-moving," the system can explain why it suggests a specific outfit combination based on current demand predictions and style logic. This transparency builds trust with store owners, enabling them to confidently optimize their inventory, reduce waste, and improve revenue in a way that a non-AI system simply could not

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achieve. By making the GLM the central engine of the product, we ensure the system doesn't just present data, but actually guides the user toward smarter, data-informed economic outcomes.

4.3.2. **Prompting Strategy:** Our approach employs a multi-step agentic prompting strategy to enhance automation in managing a clothing store. This process involves data interpretation, strategic reasoning, and natural language explanation, enabling Z.AI's GLM to function as a central intelligence hub. The strategy is effective due to the complex variables involved, as it allows for context-aware reasoning. First, the agent identifies slow-moving stock through structured sales data, then cross-references it with fashion trends to provide outfit combinations as an "AI Stylist." Finally, the system offers explanations of its decisions, fostering clarity for store owners regarding recommendations and insights based on the agentic reasoning layer, essential for insightful decision-making.

4.3.3. **Context & Input Handling**: To handle the diverse range of information inherent in the fashion industry—such as lengthy customer trend reports, social media transcripts, or extensive inventory lists—our system utilizes a sophisticated preprocessing pipeline. Since Z.AI's GLM serves as the central hub for interpreting both structured and unstructured data, we have set a maximum input limit of 128,000 tokens per session to ensure high-speed performance and context-aware reasoning. For inputs that exceed this limit, the system does not simply reject the data; instead, it employs a smart chunking and summarization strategy. Specifically, oversized text is broken down into manageable segments, where the GLM identifies and extracts the most relevant "style signals" or "sales patterns" from each chunk before recombining them into a concise, high-density prompt. This ensures that even when processing massive amounts of data, the system can still generate meaningful insights and recommend clear actions without losing the critical context needed to optimize store outcomes.

4.3.4. **Fallback & Failure Behavior:** To maintain the reliability of the CloThinks for decision-making, a Validation and Feedback Loop is implemented to address off-topic or

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incorrect responses. The Z.AI GLM generates responses that undergo evaluation by a Self-Correction Layer. If a generated strategy is flawed, the system employs an automatic retry, adhering to stricter constraints. Should the model fail consistently, it defaults to a "Simplified Insight," providing safe messaging and allowing human intervention for review. This approach protects economic outcomes while promoting effective insights and decision-making.

5. User Stories & Use Cases

- **User Stories:** *Example:* "As a product manager, I want to enter a prompt using my own way of describing so that it provides me with a design without manual drawing by me."
  User story 1: As a fashion retailer, I want to identify slow-moving stock and receive AI-suggested pairing combinations with trending items, so it can provide me with information spontaneously. Besides, I want to analyze unstructured fashion trend data so that I can predict future demand and restock inventory that aligns with market interests.

  User story 2: As a fashion-conscious shopper, I want to receive personalized outfit recommendations based on current local trends and my style preferences, so that I can make confident purchasing decisions.

- **Use Cases (Main Interactions):**

  Prompt to UI Flow:

  Users enter a natural language prompt describing a business goal (e.g., “Show me how to sell these leftover summer skirts for the upcoming fall season”). CloThinks parses the intent, identifies the specific slow-moving inventory from the database, and then renders a visual styling board on a customizable 2D Canvas that pairs those skirts with trending autumn items.

  Flow Generation:

Prompt-to-Insight Flow

User enters a query → system analyzes inventory data → AI generates insights and recommendations.

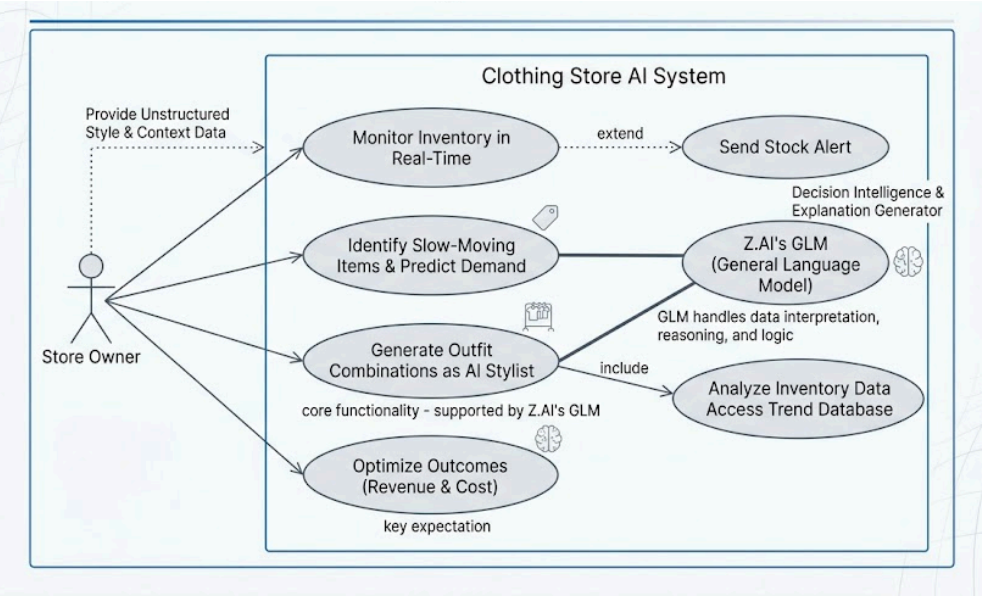
Outfit Generation Flow

User selects item → AI matches with other inventory → system generates outfit combinations.

Demand Prediction Flow

System analyzes sales trends → predicts demand → suggests restock or discount actions.

- **Use Case Diagram:**



6. Features Included (Scope Definition)

- AI-powered inventory analytics dashboard

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- One-click export to CSV and PDF summary
- Coherent and consistent design flow for entire system with text-based prompts
- Outfit recommendation engine from existing stock
- Demand prediction and stock alerts
- Customer personalized style and size recommendation support

7. Features Not Included (Scope Control)

- Advanced image-based recognition (future phase)
- Data is uploaded manually without backend integration with live POS or ERP systems
- Multi-users inputs simultaneously are not supported

8. Assumptions & Constraints

- **LLM Cost Constraint:** Estimated cost per session: ~$0.09 (23K tokens). Cost-saving design decision: Caching repeated queries for 1 hour, reducing redundant inference by ~40%.
- **Data Availability Assumption:** The system assumes that retailers maintain consistent and accurate inventory and sales data for effective analysis and predictions.
- **Technical Constraints:** The system primarily supports text-based interaction and structured inventory data; advanced visual AI features are limited in the initial version.
- **Performance Constraints:** Large datasets may slightly increase response time due to longer processing and analyzing required.
- **User Input:** System depends on accurate inventory data input and user prompts, while outputs require human validation before execution

9. Risks & Questions Throughout Development

- **Design Similarity:** How do we ensure that the AI does not generate identical inventory and styling recommendations for different retailers, which would lead to a loss of market uniqueness among SMEs?

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- **Recommendation Accuracy:** How are we ensuring outfit suggestions align with real customer preferences and trends?
- **User Adoption:** Will SME retailers adapt to AI-driven decision-making over traditional intuition?
- **Data Quality Dependency:** How to handle inaccurate or incomplete inventory data affecting AI outputs?
- **Frame Stability:** How do we ensure the AI maintains consistent and logical reasoning over time, avoiding erratic inventory suggestions that could confuse business owners or lead to fragmented procurement strategies?
- **Drawing Interpretation**: What will happen if the system misinterprets the relationship between structured inventory data and unstructured fashion trends, leading to a "hallucinated" outfit combination that pairs a high-demand item with a slow-moving item that is physically incompatible or aesthetically mismatched?

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